TZU-HSUAN CHAN

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SELF-INTRODUCTION

My previous research fields are mathematical biology and dynamical systems, which has strengthened my problem-solving skills. I enjoy learning new methods and hope to use my mathematical training to work across disciplines, particularly in physics and other fields. I would like to learn more about physics and explore how my background in mathematics could contribute to research in the area.

EDUCATION

National Yang Ming Chiao Tung University (Master's degree) *Sep. 2025 - Present*
Department of Applied Mathematics (Advisor: Chang-Hong Wu) *GPA: 4.24 / 4.30*

National Taiwan Normal University (Bachelor's degree) *Sep. 2021 - Jun. 2025*
Department of Mathematics with Scientific Computation Program *GPA: 4.05 / 4.30*

PUBLICATIONS

Extinction and the Global Structure of Positive Equilibrium in a Ratio-Dependent Intraguild Predation Model
Tzu-Hsuan Chan, Chang-Hong Wu, arxiv: https://arxiv.org/abs/2610.23995
In this work, we investigate extinction and coexistence in a three-species ratio-dependent intraguild predation model. We also derive sufficient conditions for extinction and used bifurcation analysis to characterize the emergence and global structure of the coexistence equilibrium branch.

Asymptotic Type Proportions in Topological and Random Spread Models: The Role of Observation Intervals
Jung-Chao Ban, Tzu-Hsuan Chan, Jyy-I Hong, Chen-Han Hsu, Cheng-Yu Tsai, and Nien-Cheng Tsai (under review)
In this work, we show how to choose a starting type and group generations to find long-term type proportions in spread models with fixed rules. Counting across full cycles also gives stable proportions in periodic random models.

Stability and Memory-induced Transient Oscillatory Dynamics in a Fractional HIV Model with Dual Transmission Routes
Chang-Yuan Cheng, Shyan-Shiou Chen, Tzu-Hsuan Chan, 2026, doi:10.11650/tjm/260106
In this work, we investigate the stability of a fractional-order HIV model with dual transmission routes, focusing on the role of memory effects in transient oscillatory dynamics.

HONER AND AWARDS

CTCI Foundation Outstanding Undergraduate Award in Mathematics（中技社數學傑出大學生獎） *2025*

College Student Research Creativity Award（大專學生研究創作獎） *2024*

RESEARCH EXPERIENCE

National Center for Theoretical Sciences Undergraduate Summer Researcher *Jul. 2024 - Aug. 2024*
Title: Equivalence Structures of the Spread Models (Advisor: Jung-Chao Ban, Jyy-I Hong)
Focused on exploring spread models using dynamical systems and matrix theory, deriving conditions for the existence of spread rates and model equivalence. [Report]

National Center for Theoretical Sciences Undergraduate Summer Researcher *Jul. 2023 - Aug. 2023*
Title: Curvature Equation with Conical Singularities and Integrable Lamé Equation (Advisor: Ting-Jung Kuo)
Analyzed the curvature equation using Liouville's theorem, the Lamé equation, and elliptic functions, deriving necessary and sufficient conditions for solution existence. [Report]

SKILLS

Differential Equations, Dynamical Systems, Matlab, Modeling, LaTeX

TEACHING EXPERIENCE

Calculus and Partial Differential Equation *NYCU*
Graded quizzes and homework.
Calculus, Advanced Calculus, Applied Mathematics, and Differential Equations *NTNU*
Prepared and graded quizzes and homework, and led TA sessions.